Power Series and Taylor Series

Section 1: Radius and Interval of Convergence

1 Find the radius of convergence R for the power series $\sum_{n=1}^{\infty} \frac{(x-2)^n}{n \cdot 3^n}$.

a. $R = 3$

b. $R = 1/3$

c. $R = 1$

d. $R = \infty$

2 Find the exact interval of convergence for $\sum_{n=1}^{\infty} \frac{(x-2)^n}{n \cdot 3^n}$.

a. $[-1, 5)$

b. $(-1, 5)$

c. $[-1, 5]$

d. $(-1, 5]$

3 Find the radius of convergence of $\sum_{n=0}^{\infty} n! \, (2x)^n$.

a. $R = 0$

b. $R = 1/2$

c. $R = 1$

d. $R = \infty$

Section 2: Geometric Series Representations

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4 Find the power series representation for $f(x) = \frac{1}{1+2x}$ centered at 0.

a. $\sum_{n=0}^{\infty} (-1)^n 2^n x^n$

b. $\sum_{n=0}^{\infty} 2^n x^n$

c. $\sum_{n=0}^{\infty} (-1)^n x^n$

d. $\sum_{n=1}^{\infty} (-2x)^n$

5 Find the Maclaurin series for $\frac{x}{4-x^2}$.

a. $\sum_{n=0}^{\infty} \frac{x^{2n+1}}{4^{n+1}}$

b. $\sum_{n=0}^{\infty} \frac{x^{2n}}{4^n}$

c. $\sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{4^{n+1}}$

d. $\frac{1}{4}\sum_{n=0}^{\infty} x^{2n+1}$

6 Find the Maclaurin series for $\ln(1+x)$ by integrating the series for $\frac{1}{1+x}$.

a. $\sum_{n=1}^{\infty} (-1)^{n-1} \frac{x^n}{n}$

b. $\sum_{n=0}^{\infty} (-1)^n \frac{x^{n+1}}{n+1}$

c. Both of the above are equivalent

d. $\sum_{n=1}^{\infty} \frac{x^n}{n}$

Section 3: Standard Maclaurin Series

7 State the Maclaurin series for e^x.

a. $\sum_{n=0}^{\infty} \frac{x^n}{n!}$

b. $\sum_{n=1}^{\infty} \frac{x^n}{n}$

c. $\sum_{n=0}^{\infty} (-1)^n \frac{x^n}{n!}$

d. $\sum_{n=0}^{\infty} \frac{x^{2n}}{(2n)!}$

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8 State the Maclaurin series for $\cos(x)$.

a. $\sum_{n=0}^{\infty}(-1)^n\frac{x^{2n}}{(2n)!}$

b. $\sum_{n=0}^{\infty}(-1)^n\frac{x^{2n+1}}{(2n+1)!}$

c. $\sum_{n=0}^{\infty}\frac{x^{2n}}{(2n)!}$

d. $\sum_{n=1}^{\infty}(-1)^n\frac{x^{2n-1}}{(2n-1)!}$

9 Find the first three non-zero terms of the Maclaurin series for $\sin(2x)$.

a. $2x-\frac{4}{3}x^3+\frac{4}{15}x^5$

b. $2x-4x^3+\frac{4}{3}x^5$

c. $2x-\frac{8}{6}x^3+\frac{32}{120}x^5=2x-\frac{4}{3}x^3+\frac{4}{15}x^5$

d. Both of the above are correct

Section 4: Taylor Polynomials and Lagrange Error Bound

10 Find the 2nd-degree Taylor polynomial $T_2(x)$ for $f(x)=\sqrt{x}$ centered at $a=4$.

a. $T_2(x)=2+\frac{1}{4}(x-4)-\frac{1}{64}(x-4)^2$

b. $T_2(x)=2+\frac{1}{4}(x-4)-\frac{1}{32}(x-4)^2$

c. $T_2(x)=2+\frac{1}{2}(x-4)-\frac{1}{16}(x-4)^2$

d. $T_2(x)=2+\frac{1}{4}(x-4)+\frac{1}{64}(x-4)^2$

11 State the formula for the Lagrange Error Bound for a Taylor polynomial $T_n(x)$ centered at a.

a. $|R_n(x)|\le\frac{M}{(n+1)!}|x-a|^{n+1}$, where $M=\max|f^{(n+1)}(t)|$ on $[a,x]$

b. $|R_n(x)|\le\frac{M}{n!}|x-a|^n$

c. $|R_n(x)|\le M|x-a|^{n+1}$

d. $|R_n(x)|\le\frac{f^{(n+1)}(a)}{(n+1)!}|x-a|^{n+1}$

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12 Use the 3rd-degree Maclaurin polynomial for e^x to bound the error in estimating $e^{0.1}$. (Take $e^{0.1} < 3$).

a. $|R_3(0.1)| \le \frac{3}{24}(0.1)^4 = 1.25 \times 10^{-5}$

b. $|R_3(0.1)| \le 10^{-4}$

c. $|R_3(0.1)| \le \frac{1}{6}(0.1)^3$

d. $|R_3(0.1)| \le 10^{-6}$

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